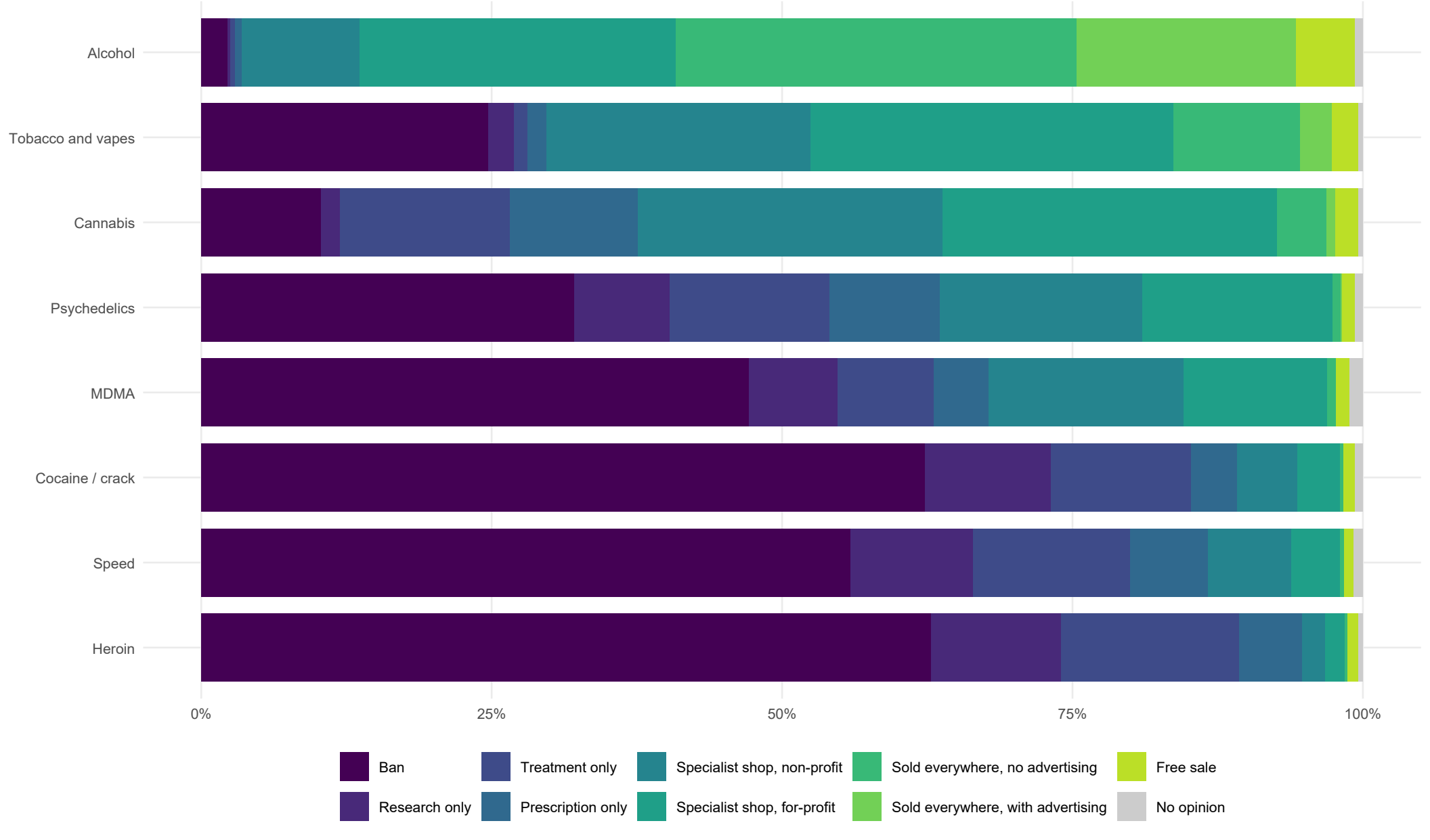
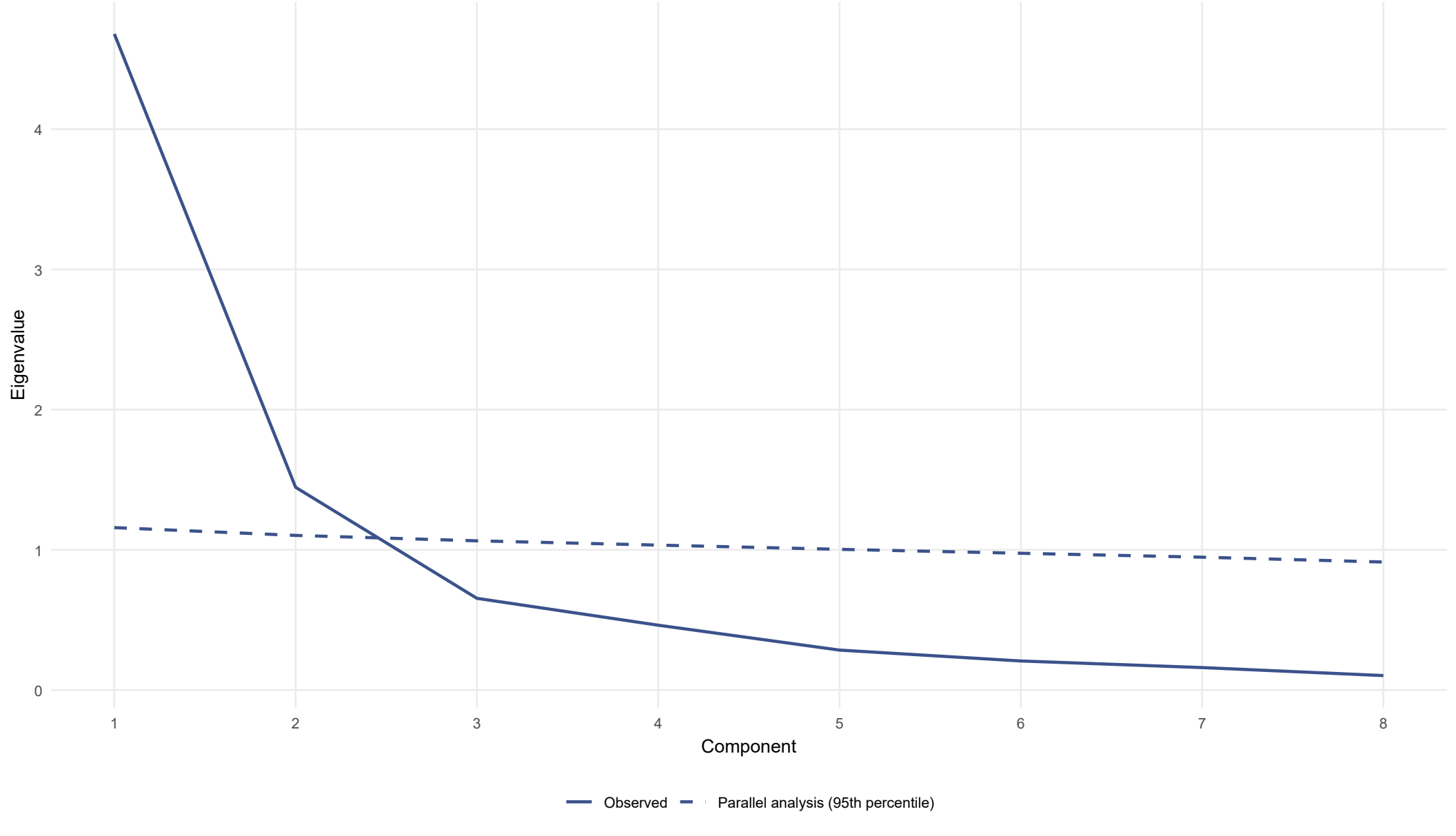
Step 1. Answer distributions (unweighted)

Nine ordered, skewed options with sparse categories (and empty ones for the hard drugs): treat the items as ordinal, with polychoric correlations and WLSMV



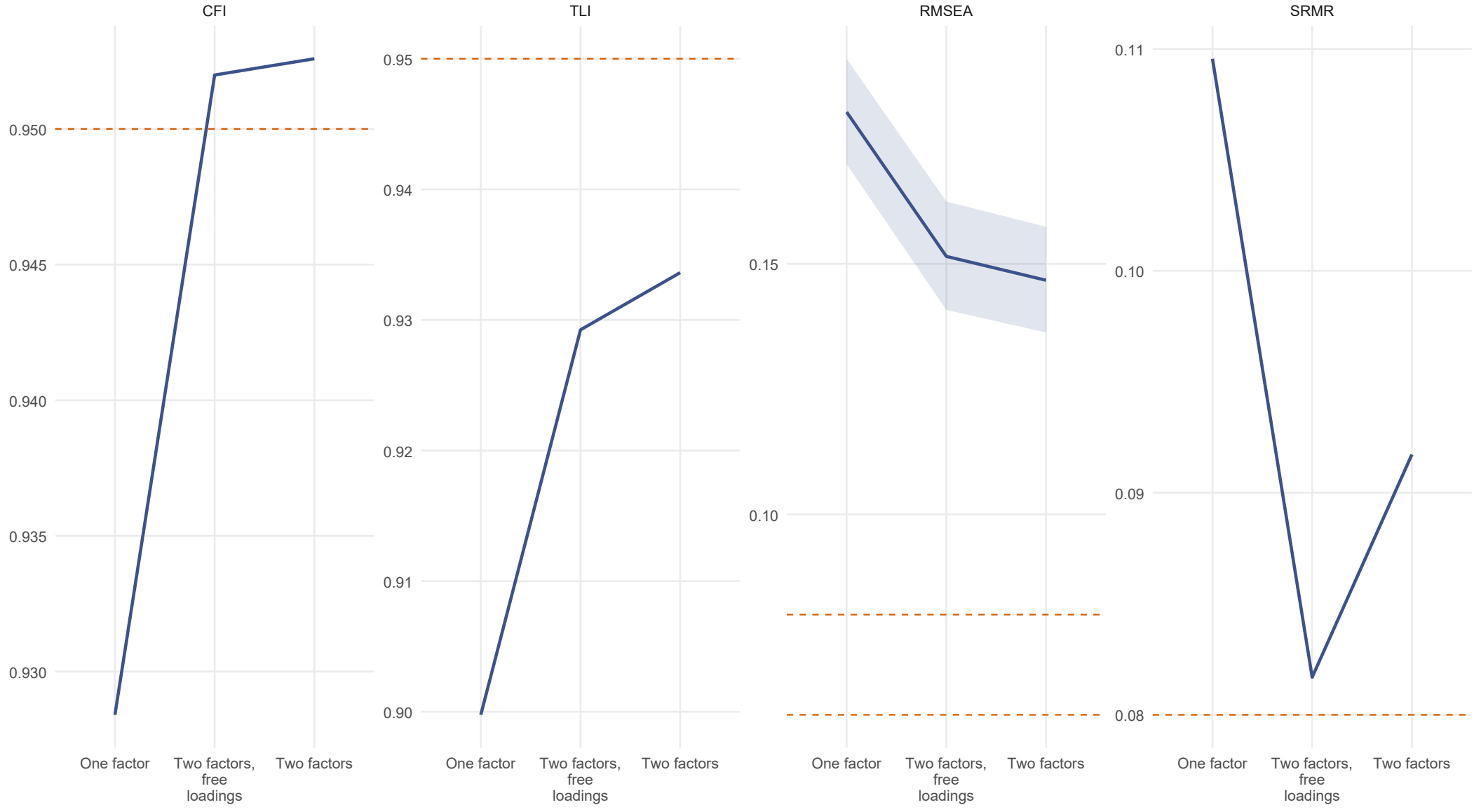
Step 2. Scree plot with parallel analysis

Eigenvalues of the weighted polychoric correlations. Factors to retain: observed eigenvalues above the parallel-analysis line.
One dominant factor and a clear second one: test a licit and an illicit factor against one factor



Step 3. One factor or two

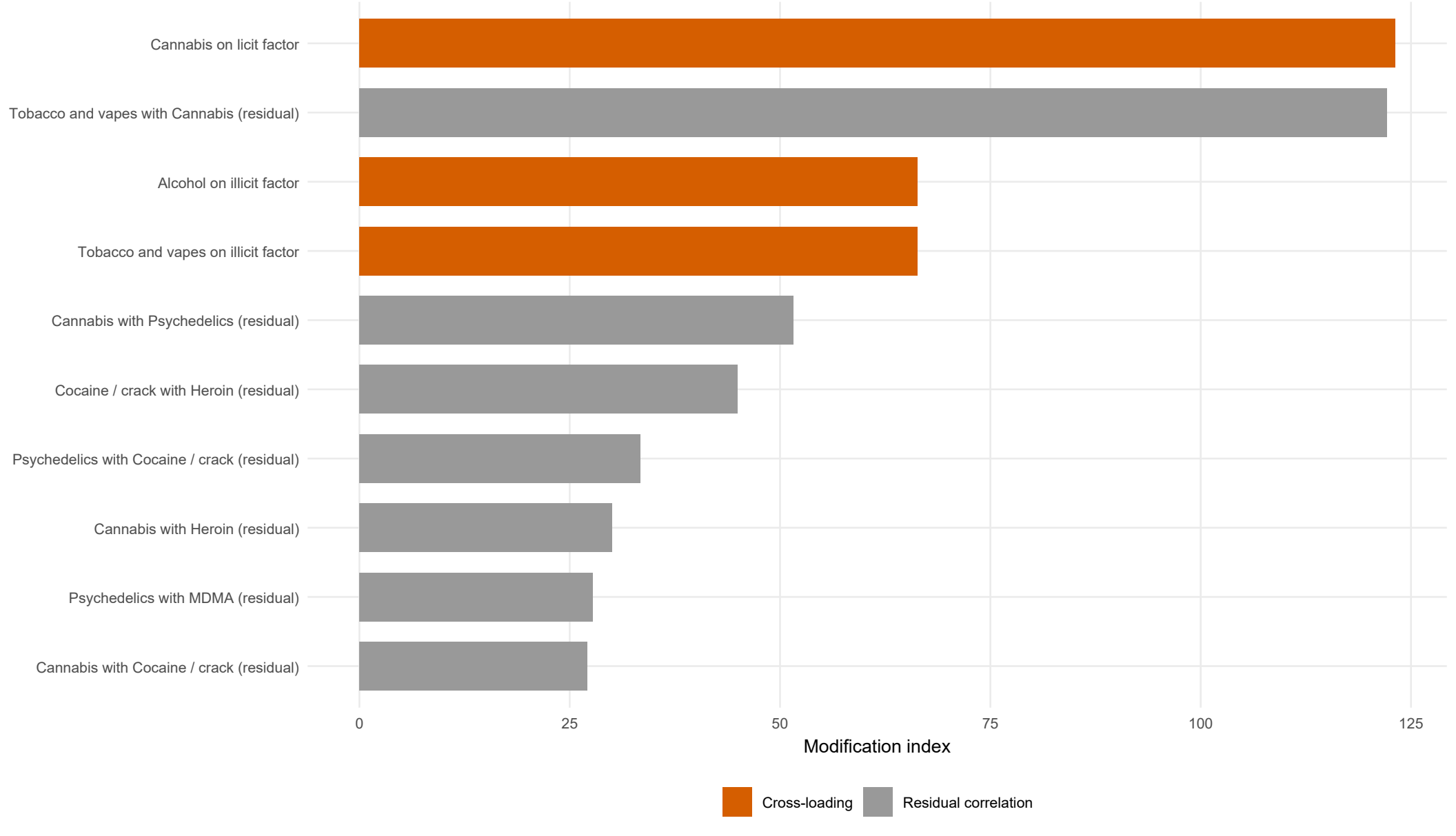
Two factors fit clearly better (scaled chi-square difference test). The free licit factor gives tobacco a negative residual variance, so its two loadings are set equal; the fit stays poor (RMSEA), so look at where the misfit is



Dashed: conventional cut-offs (Hu & Bentler 1999). Band: 90% CI of the RMSEA.

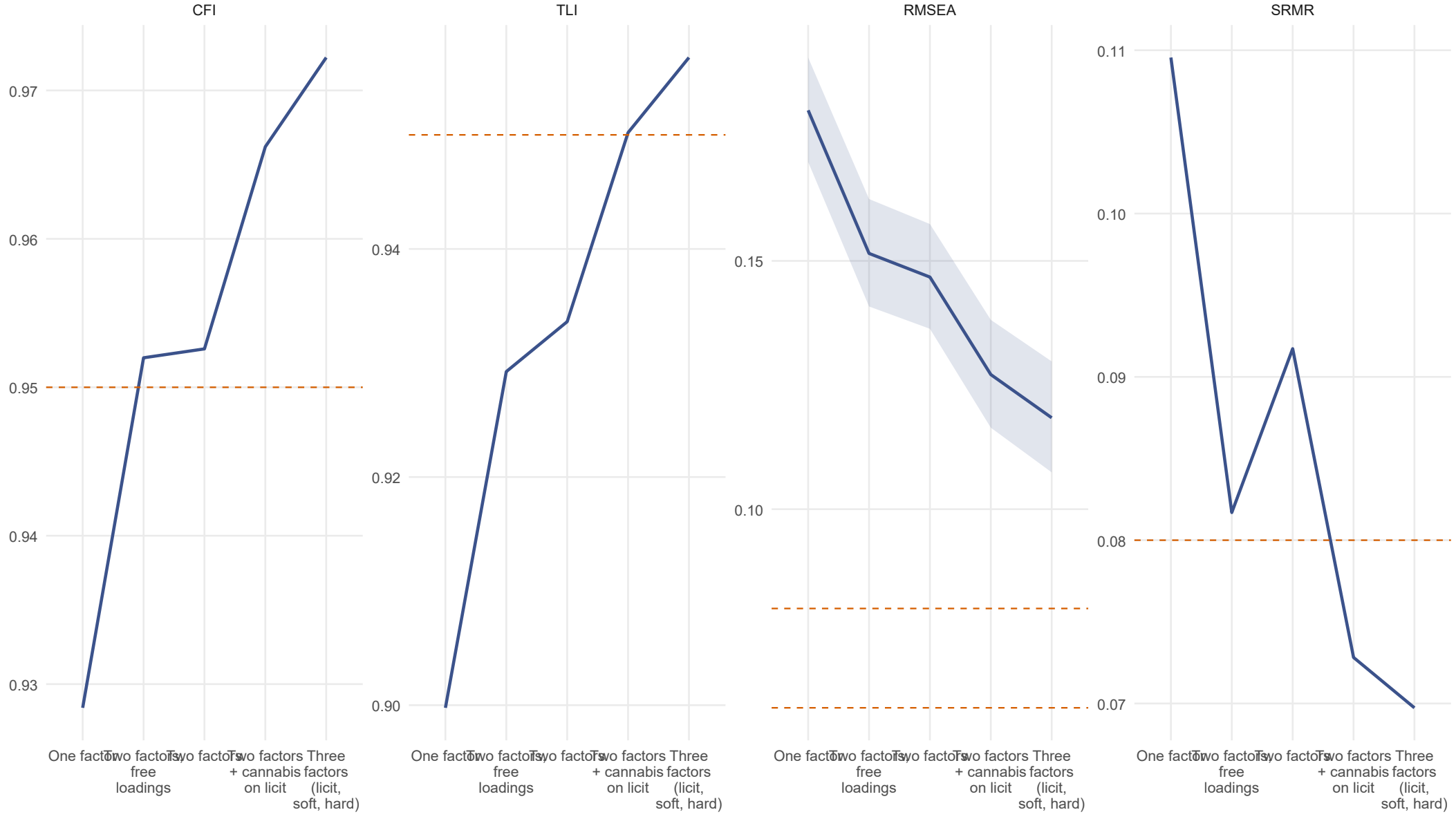
Step 4. Modification indices, two-factor model

Largest misfit: cannabis also loads on the licit factor. Plausible, as cannabis is tolerated and sold in coffeeshops: free that loading.
The residual correlations among cocaine, speed and heroin point to a soft/hard split within the illicit drugs



Step 5a. Fit of all models

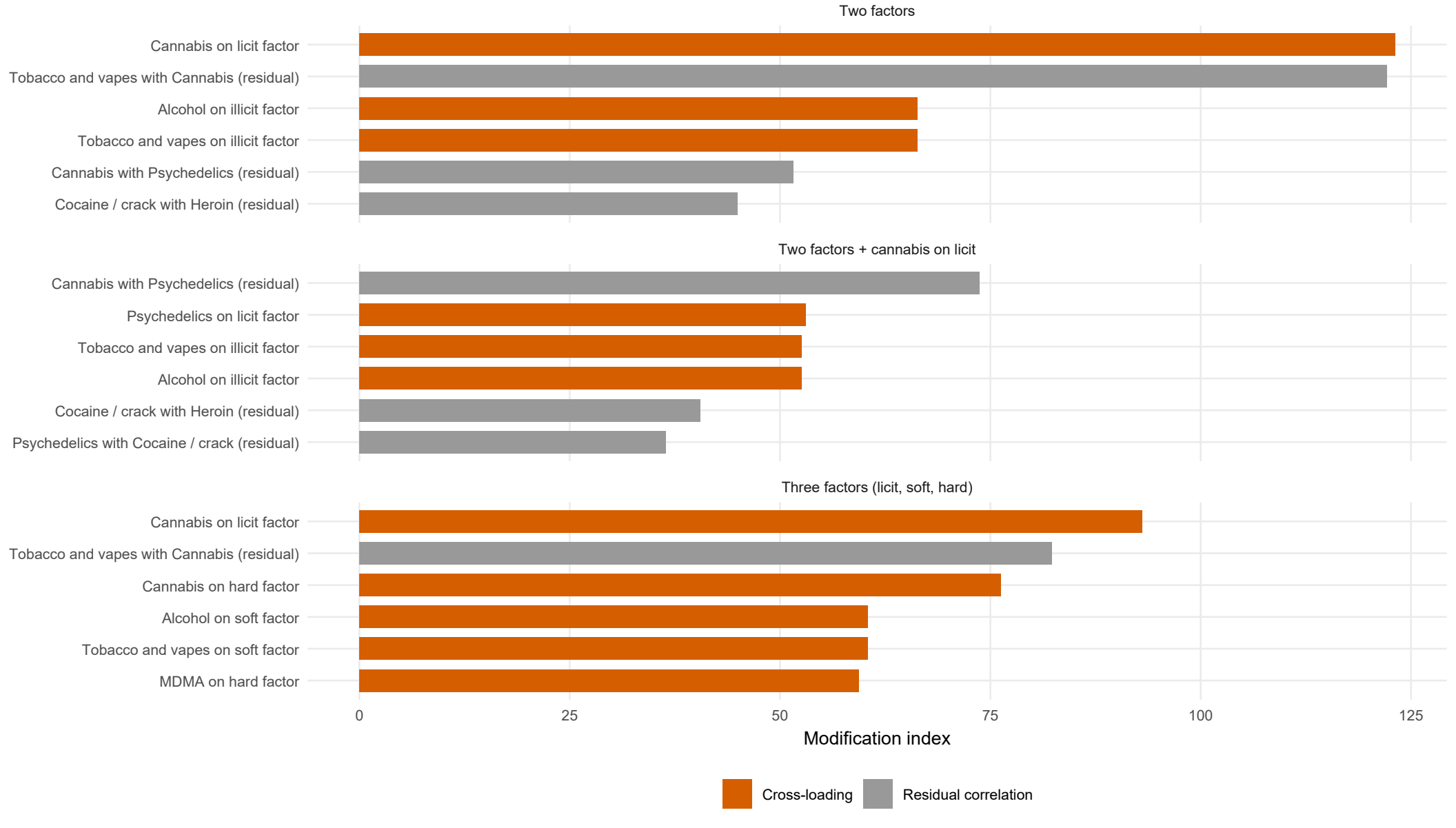
Both the cannabis cross-loading and the soft/hard split improve the fit; the licit-illicit correlation is what matters, so compare it across models



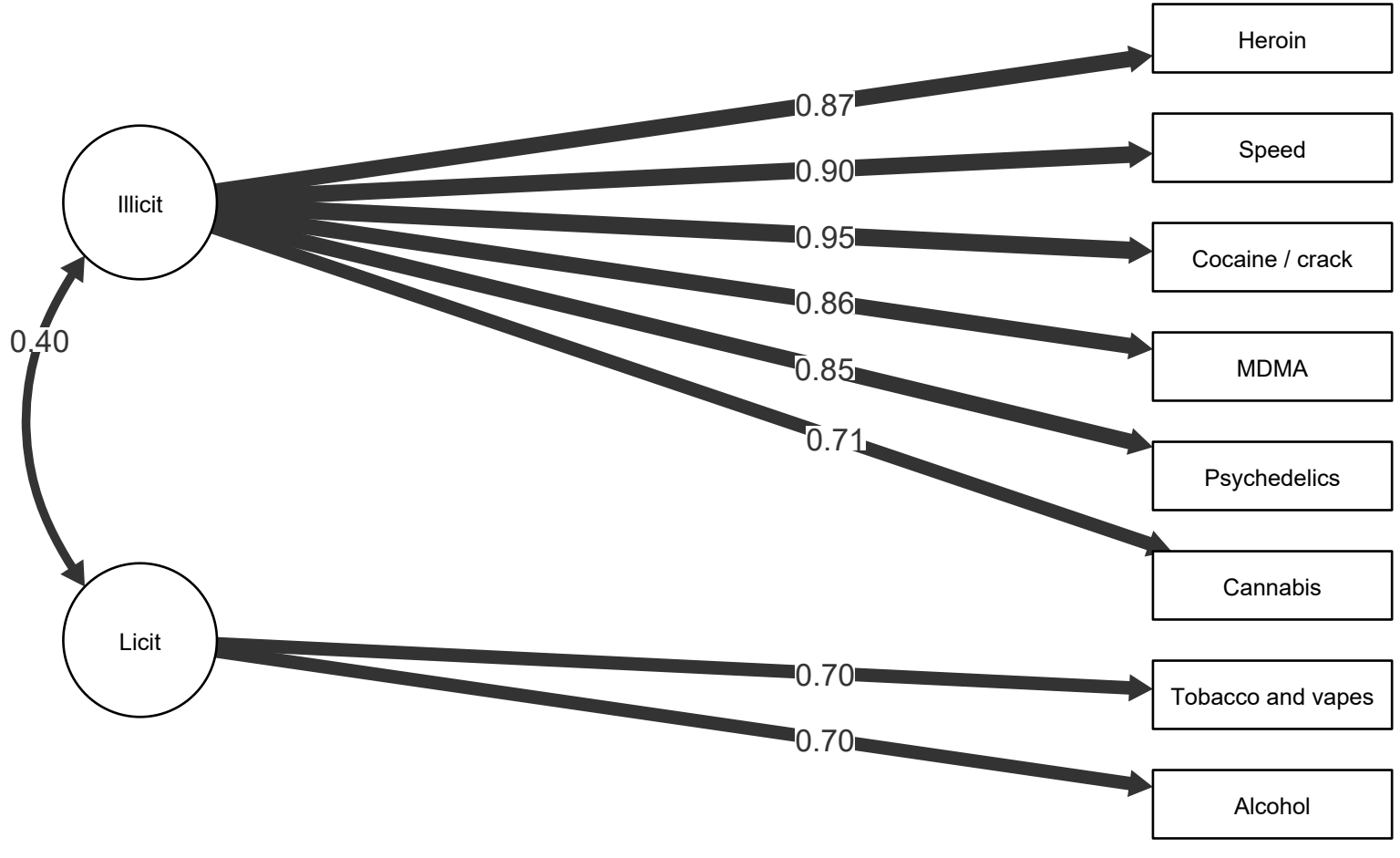
Dashed: conventional cut-offs (Hu & Bentler 1999). Band: 90% CI of the RMSEA.

Step 5b. Largest remaining modification indices

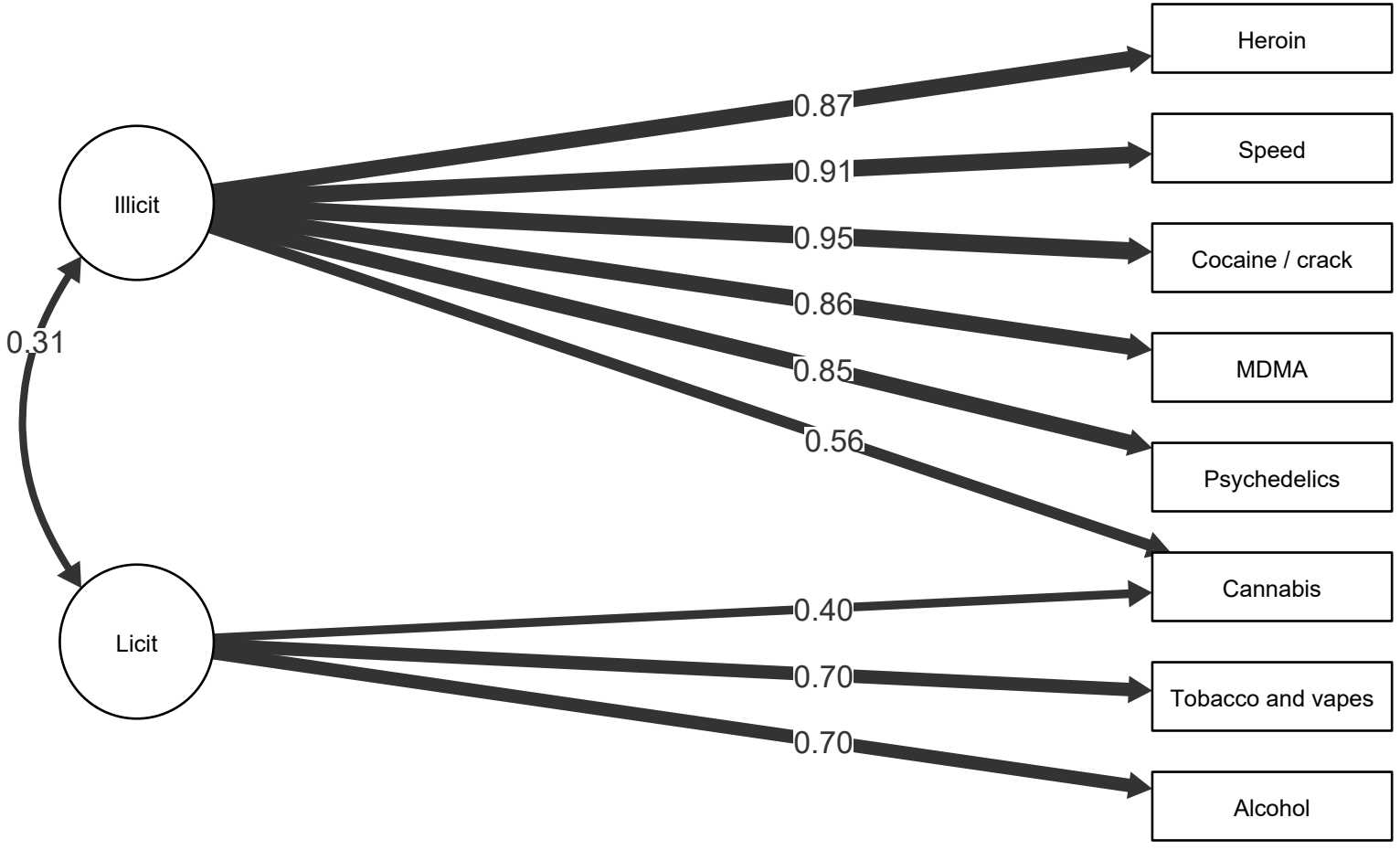
In the two-factor model the cross-loading takes out the cannabis misfit. The soft/hard split takes out the hard-drug residuals, but there cannabis still wants to load on licit: cannabis sits between the licit and the illicit drugs



Step 6a. Two-factor model, standardised estimates



Step 6b. Two-factor model with cannabis on licit, standardised estimates



Step 6c. Three-factor model, standardised estimates

